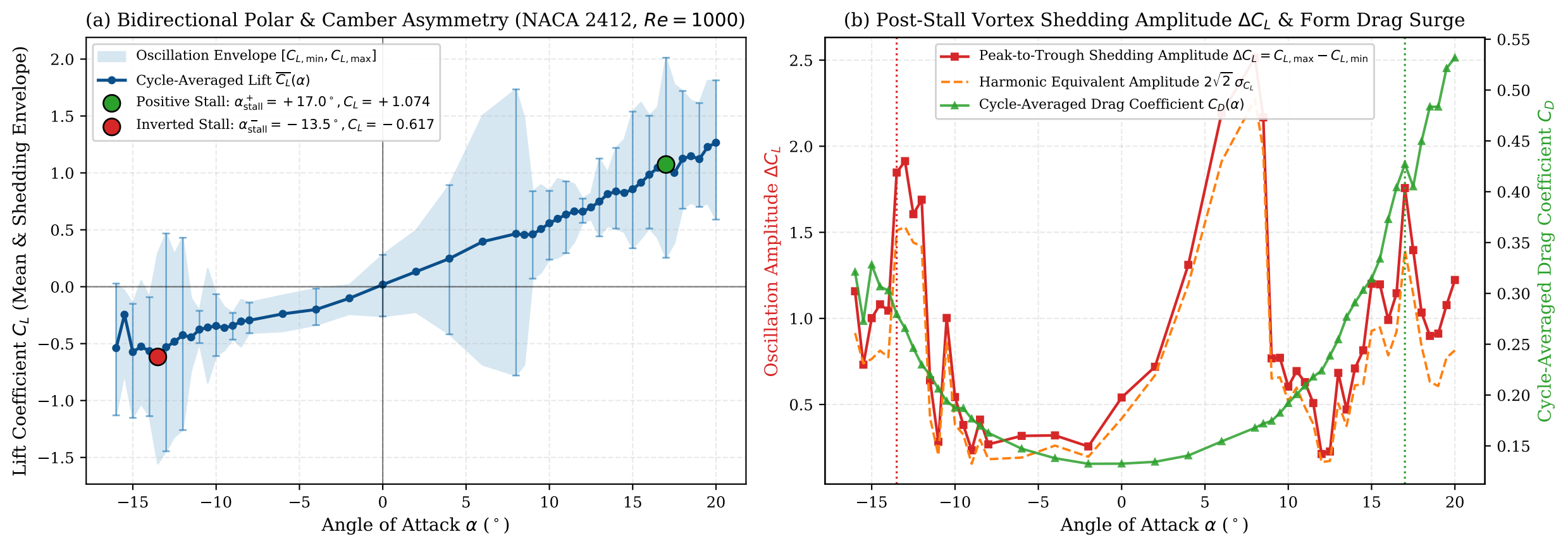
(a) Bidirectional Polar & Camber Asymmetry (NACA 2412, $Re = 1000$)
Oscillation Envelope $[C_{L,\min}, C_{L,\max}]$
Cycle-Averaged Lift $\overline{C_L}(\alpha)$
Positive Stall: $\alpha^{+}_{\text{stall}} = +17.0^\circ, C_L = +1.074$
Inverted Stall: $\alpha^{-}_{\text{stall}} = -13.5^\circ, C_L = -0.617$
Lift Coefficient C_L (Mean & Shedding Envelope)
Angle of Attack α ($^\circ$)
(b) Post-Stall Vortex Shedding Amplitude ΔC_L & Form Drag Surge
Peak-to-Trough Shedding Amplitude $\Delta C_L = C_{L,\max} - C_{L,\min}$
Harmonic Equivalent Amplitude $2\sqrt{2}\,\sigma_{C_L}$
Cycle-Averaged Drag Coefficient $C_D(\alpha)$
Oscillation Amplitude ΔC_L
Cycle-Averaged Drag Coefficient C_D
Angle of Attack α ($^\circ$)